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Options available for speech encryption

Professor H. J. BEKER, BSc, BA, PhD, AFIMA, MIS*

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SUMMARY

This paper presents two of the options available for speech encryption. The first is an analogue technique utilizing the principle of time element scrambling but introducing the concept of a sliding window. It is shown that this enables the system time delay to be reduced while simultaneously increasing the number of scrambling permutations available to the user. The second option is a digital technique using a stream cipher. Some of the requirements for such a system are discussed. Finally, the author attempts to show where these two options fit into the range of currently available techniques, from a security viewpoint.

* *Racal-Comsec Limited; now with Racal Research Ltd, Worton Drive, Worton Grange Industrial Estate, Reading, Berkshire RG2 0SB*

1 Introduction

Speech is probably the most fundamental form of communication available to us, and our society has become highly dependent on our modern, fast and accurate means of transmitting spoken messages. Usually the main aim of communicators is merely to transmit a message as quickly, accurately and cheaply as possible. There are, however, a number of situations where the information is confidential and where an interceptor might be able to benefit immensely from the knowledge gained by monitoring the information circuit. In such situations, the communicators must take steps to conceal and protect the content of their spoken message. Of course, the amount of protection will vary. On occasions it is sufficient to prevent the casual listener from understanding the message, but there are other times when it is crucial that even a determined interceptor must not be able to deduce its content.

One basic problem facing the designer of speech security equipment is that there already exists a great variety of communications circuits for the transmission of speech signals. Many of the techniques at his disposal might necessitate the restriction of the communicator to only a few of these types of channel. Furthermore, the designer must remain aware at all times that almost all speech security systems reduce, at least to some extent, the audio quality of the voice transmission. Clearly security will not be enhanced if the link has been so badly degraded that the same message has to be repeated a number of times. There is therefore a need to take into consideration the type of transmission link that might be used, and to choose for any particular security equipment an encryption system that will give the least degradation of audio quality. There is no point in having a very high security level if it is no longer possible to communicate!

The object of this paper is to introduce two new options for speech security. The first is based on an analogue transmission of the signal, the second on a digital transmission. Each of these options is discussed and their respective security levels, advantages and disadvantages are considered.

The difference between analogue and digital cipher systems should first be noted. Starting with the analogue system on the transmit side, the voice input is converted into a serial data stream using an analogue to digital converter. It is then scrambled, and after scrambling re-converted to an analogue signal for transmission. There are a number of important points to notice here. First, the signal being transmitted is actually an analogue signal and will therefore contain some of the characteristics of a speech signal. The a-d and d-a converters are really there just for convenience of processing. The system is a truly analogue one in that it starts, and ends, with an analogue signal. The idea behind the scrambler is to preserve, as nearly as possible, the original speech bandwidth of the signal, so the final output analogue signal occupies as nearly as possible the same bandwidth as the input signal. This clearly has a number of advantages and in particular it means that the system is essentially independent of the kind of communication channel involved. For instance, this system can be used over narrow-band h.f. links or over telephone lines.

The concept of a digital cipher system is in some ways simpler. Here again the voice input is converted, using an analogue to digital converter, into a serial data stream. This is then directly enciphered. In transmitting the resulting binary digits, which it is hoped will look like random binary digits, the characteristics of the original voice signal will have been completely destroyed. However, with such a system there is the possible disadvantage that unless the a-d converter is a sophisticated low-bit-rate voice coder, the signal that results may actually occupy a bandwidth that is larger than that of the input signal and too large, in the case of some communications circuits, for the particular channel to be used.

2 An Analogue Cipher Technique

In introducing the analogue cipher technique to be described, it is convenient to consider first the so-called fixed window time element scrambler. As shown in Fig. 1, this entails dividing the speech input signal into 'frames of speech'. Each frame of speech is then sub-divided into a

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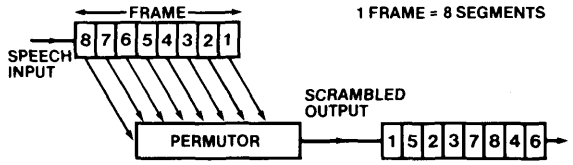


Fig. 1. Fixed window time element scrambling

number of segments. In this example one frame of speech is divided up into eight segments. The idea then is in some way to scramble the order of these segments; in other words, transmit them in a different order from that in which they were received. The term 'window' refers to the time over which any given segment may be displaced. In this particular example, segment 6 is transmitted first, then segment 4, then segment 8 and so on. Clearly in using a system like this there are a number of parameters to be decided on. For instance, the choice of permutations to be used—are they all good, or are there some that will leave more 'residual intelligibility' in a signal than others? (The residual intelligibility of an output signal is that proportion of the original signal which can be understood directly when listening to the enciphered message.) The duration of the frame, or frame length, has to be considered, and also the length of each particular segment.

Taking the segment length first, there are two conditions that dominate. The first is the need for the segments to be as short as possible, since there is no scrambling actually within the segments. For instance, if segments were longer than a spoken word this clearly would give a lot of residual intelligibility after scrambling. Unfortunately, however, reducing the duration of the segments tends to reduce audio quality in the recovered signal, because more and more boundaries (i.e. discontinuities) are thereby created in the speech signal. As a compromise between these conflicting requirements the segment length tends to vary between about 20 ms and 60 ms in typical equipments available on the market today.

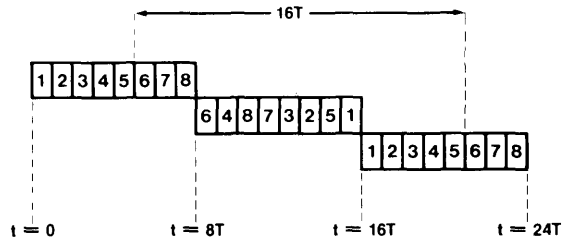


Fig. 2. Timing diagram for fixed window scrambler. Segment duration = T seconds

As to frame length, the requirements may be better understood by considering a timing diagram such as Fig. 2. Assuming a segment duration, or segment period, of T seconds, the diagram begins at time $t = 0$ when the first segment of a frame is spoken into the machine. Clearly it will take up to $t = 8T$ to complete the entering of the first frame. Now, if any permutation is to be allowed, transmission cannot start until time $t = 8T$ because (although not the case in this example) it might be desired to transmit segment 8 first and this could not be done until it had been received. Only at time $t = 8T$ therefore, can the transmission begin. Similarly at the receive end, as does happen in this example, it may be that the first segment to be output is the last one to be received, and it would not be until time $t = 16T$ that output of the recovered signal could start. Looking at any particular segment, say segment 6, even allowing negligible time for transmission, it can be seen that there is a delay of $16T$ seconds. (In general, with M segments per frame the delay time will be $2MT$.)

The effect of this delay is similar to that experienced on international telephone calls which are transmitted via satellite. From the user's point of view it is undesirable and so, in deciding on the frame length, the first requirement is that the frame should be as short as possible in order to minimize this delay. On the other hand, as the frame length is made shorter, this is going to reduce the number of permutations available. In general, with a frame of M segments, the number of permutations available is $M!$, $(M \times (M-1) \times \ldots + 1)$. Clearly as the frame length becomes shorter the number of different permutations available also becomes smaller. Again unfortunately these are conflicting requirements. Between 8 and 16 segments per frame are typical in equipments available at the moment.

The permutations may be considered in relation to the 8 segments-per-frame example, in which 8! permutations are available, representing 40 320 different possible permutations on the 8 elements. Unfortunately many of those permutations leave a certain amount of residual intelligibility in the signal; with some permutations, even after scrambling, a great deal of the signal can still be understood. It is therefore necessary to look at these permutations and decide which leave a sufficiently low residual intelligibility and which leave a high residual intelligibility making them unacceptable. Many authors who have written on this subject seem to agree that out of the 40 320 in this example, only about 3000 would provide a sufficiently low residual intelligibility in the finally scrambled signal.

3 The Sliding Window System

Turning now to what is known as the sliding window type of time element scrambler, its aim is to reduce the delay imposed by the fixed window type and simultaneously to increase the number of permutations. The method used involves restricting the type of permutations that can be used for the scrambling, specifically to permutations in the set $A(n, k)$ where this is a set of permutations which guarantee unequivocally that no segment will be delayed for too long. This set $A(n, k)$ has two parameters—n and k. The n refers to the number of segments per frame in the same way as for the fixed window. The parameter k refers to the maximum amount of time that any segment will be delayed, and only those permutations will be used which guarantee that no segment will be delayed within the system for more than k segment periods. Thus in this case the 'window' can be said to be sliding. More precisely, the set $A(n, k)$ is the set of permutations, α (in S_n) such that the image of any element under α belongs to the set $\{\overline{i-1}, \ldots, \overline{i-k}\}$ where S_n is the set of all permutations on n elements and the bar denotes the residue class modulo n with the classes labelled $1, 2, \ldots, n$.

As an example, using parameters small enough for convenient illustration (Fig. 3), let $n = 8$ and $k = 3$.

Fig. 3. Example of a sliding window permutation

$$\begin{pmatrix} 1\,2\,3\,4\,5\,6\,7\,8 \\ 6\,1\,8\,3\,2\,5\,4\,7 \end{pmatrix}$$ belonging to A(8,3)

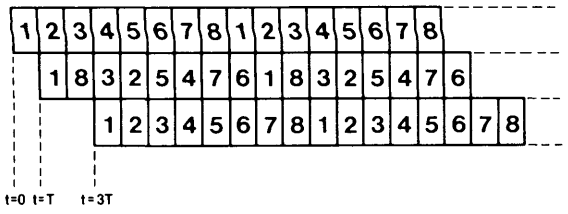


Fig. 4. Timing diagram for sliding window example

Starting with the original ordered segments 1, 2, 3 up to 8, the permutation (corresponding to a scrambled frame of speech) is written below. Now with $n = 8$, the frame size is 8 and there are 8 segments per frame. It will be remembered that the parameter 3 refers to the delay time and means that no segment here is delayed for more than 3 segment periods. Looking therefore for when segment 1 is to be transmitted it must lie under 2, 3 or 4. The transmission of segment 2 must lie under 3, 4 or 5; 3 must lie under 4, 5 or 6 and so on. The numbers 6 and 8 occurring near the beginning of the frame relate to residue classes modulo n, so in fact the 6 and the 8 refer to the appropriate segments in the previous frame of speech. Again this will become clearer on looking at a timing diagram (Fig. 4), which for clarity shows two frames of speech.

With this system, immediately the first segment is spoken into the machine, the scrambling can start and it can run simply according to the pattern shown. The order is precisely that of the permutation that has been defined. Now on the receive side, because of the permutation restriction, no segment is delayed for more than 3 segment periods. At time $t = 3T$, in other words after the duration of 3 segment periods, it can be guaranteed that any segment required will either have already arrived or will now be arriving. So, for instance, by the time 1 is required, it has already arrived; by the time 2 is required it is arriving, by the time 3 is required it has already arrived, and so on. Thus for this particular system, with $k = 3$, the delay is only $3T$ seconds. In general, for an arbitrary k, the delay time will be kT seconds. It can be seen that even from this small example the delay time has been quite dramatically decreased. From the outset, however, there were two aims: to reduce the delay time and to increase the number of permutations available. It must now be determined whether the second of these goals has been achieved.

It turns out in practice that evaluating the sizes of these sets $A(n, k)$ is a difficult problem and no general formula is known at the present time for doing this. However, in collaboration with Dr. Chris Mitchell of Racal-Comsec, an algorithm has been developed for evaluating these numbers. Some examples are given in Fig. 5 which shows how the size of the set increases for varying parameters. In fact, as these are increased, the number of permutations available increases exponentially. As a comparison, the figures shown in brackets on the right hand side are the number of permutations that would be available with the fixed window system for the same delay time. These few examples are sufficient to demonstrate how dramatically the sliding window approach can increase the number of permutations available.

Fig. 5. Examples of the number of permutations available (sliding window T.E.S.) (Algorithm due to Beker & Mitchell, Racal-Comsec Ltd., 1981)

$\lvert A(8,4) \rvert$	$= 264$	(2)
$\lvert A(24,8) \rvert$	$= 8{\cdot}75 \times 10^{12}$	(24)
$\lvert A(48,8) \rvert$	$= 3{\cdot}68 \times 10^{25}$	(24)
$\lvert A(48,12) \rvert$	$= 1{\cdot}67 \times 10^{33}$	(720)

Unfortunately, not all of these permutations are useful and some leave too much residual intelligibility. As an instance of a bad permutation, consider another example from A(8, 3), in Fig. 6. Looking at the transmit side of the timing diagram, it can be seen that there are periods where segments 3 to 8 are actually transmitted consecutively.

$$\begin{pmatrix} 1\,2\,3\,4\,5\,6\,7\,8 \\ 7\,8\,2\,1\,3\,4\,5\,6 \end{pmatrix}$$ **belonging to A(8,3)**

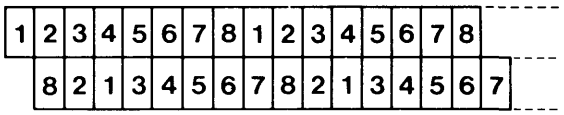


Fig. 6. Example of bad permutation

This means that in fact three-quarters of the original frame (6 segments out of 8) are transmitted in precisely the order they began. Obviously that will leave a great deal of residual intelligibility and such permutations are therefore undesirable. In experiments carried out at Racal-Comsec, it turns out that with regard to the choice of permutations, the most important requirement is that no two consecutive segments in the original frame should remain consecutive after scrambling. In other words, any pair of consecutive segments must be separated by the scrambling system. There are actually other conditions involved, but these can be neglected here.

As was stated earlier, the final signal from any analogue scrambling system still retains some characteristics of the original speech. There are a number of ways in which this might be reduced. One such method, to enhance the actual time scrambling, is to reverse the order of bits within a segment of the speech. It will be recalled that, for ease of processing, the signal undergoes an a-d conversion and then again a d-a conversion. This means that the segments of the speech are actually held within a store of the machine as a number of consecutive bits. Now there is no reason why they should be read out in the same order as they were written in; they could be read out in the reverse order for example. Better still, a key could be used to determine whether to read them out in the original order or in the reverse order. This is an improvement inherent in the system—the ability to reverse the order of bits in an element of the digital store—and it results in a reduction of about 10% in the level of residual intelligibility.

Another way of improving the system is to add a 'second dimension'. There is nothing to prevent the introduction of scrambling within the frequency domain (as opposed to the time domain so far discussed), for example, by varying the clock rate of the a-d and d-a converters. Once again a key could be used to vary this clock rate. The improvement so obtained amounts to about 15%. A third technique is to perform what can be described as full-frequency scrambling, as well as time scrambling, breaking the full frequency spectrum into sub-bands and scrambling these

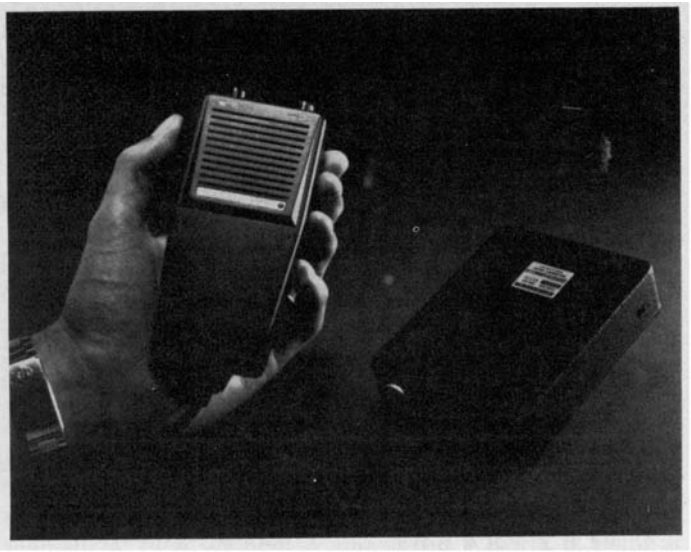


Fig. 7. Racal-Comsec MA4258 with keyfill unit

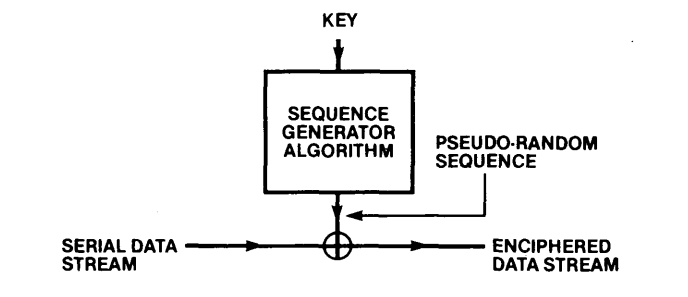
in the same sort of way as employed in the time scrambling. This leads to about a 20% improvement.

Unfortunately, these three ways of reducing the residual intelligibility are not only ordered above in terms of the improvement they yield, they are also ordered in terms of cost and complexity and in terms of the degradation they produce in the audio quality of the recovered signal. Thus, the first of these does not increase the complexity very much and does not greatly reduce the quality of the signal, whereas the full frequency scrambler does vastly increase the complexity of the equipment, and might substantially reduce the audio quality over certain types of communication channel.

An example of a device that uses the sliding window technique is shown in Fig. 7. This is the Racal-Comsec MA4258. One of the advantages of this technique is its highly attractive implementation. For convenience of key management, a keyfill unit is provided so that the key (which will be kept secret) can be entered into this unit. The keyfill unit, which contains the intelligence for generating permutations and ensuring that they will give only a small residual intelligibility, then generates the permutations from the keys and transfers them into the MA4258s. The MA4258 itself is capable of storing the permutations corresponding to up to 4 different key settings. In this way some degree of key management is incorporated within the MA4258s.

To fill in some of the parameters for this particular device, the segment length in this case is about 25 ns, the frame length n is 48. The segment delay k is actually changeable between 8, 12 and 16, and can be set depending on the user's particular requirements. These particular segment delays imply, respectively, delay times of about 200, 300 or 400 ms. Again, corresponding to these different delays, there are different numbers of permutations available. For example, with the 8 segments delay there are more than 10^{21} and the 16 segment delay offers more than 10^{38} different available permutations. The MA4258 also incorporates key-dependent segment reversals and key-dependent variations on the clock rates of the a-d and d-a converters. Thus it is a 2-dimensional scrambler with very low residual intelligibility.

Fig. 8. Stream cipher system



4 A Digital Cipher Technique

As already discussed, with a digital system the voice input is converted into a digital signal and the resulting digits are directly enciphered and then transmitted. The most advantageous cipher technique used in this type of system is a stream cipher and its operation is simple (Fig. 8). The heart of the machine is a sequence generator algorithm into which the selected key is entered. The algorithm then uses this key as a seed to produce a pseudo-random number sequence, which in this case would consist of binary digits. The serial data stream is of course the digitized voice that has been discussed earlier, and simply by modulo-2 addition (EXCLUSIVE-OR operation) the enciphered data stream is obtained.

Another commonly used technique for digital cipher systems of this kind is also to include a feedback from the enciphered data stream into the sequence generator, giving two inputs to the sequence generator: the enciphered data stream and the key. This has both advantages and disadvantages. The first of the disadvantages relates to security, in that someone intercepting the enciphered data stream will become aware of one of the inputs to the algorithm (i.e. the input fed back). In many circumstances this can reduce the security level. The second disadvantage is more important from the user's point of view, and is that in feeding back enciphered data into the algorithm, errors incurred during transmission may cause whole blocks of errors at the receive end, through what is known as 'error propagation'. The effect this has from the user's point of view is to reduce the range of his radio. These two disadvantages make the cipher feedback technique rather unattractive. Its only advantage is that the system is self-synchronizing. With the stream cipher system a separate synchronization signal has to be provided. Nonetheless, because of its more reliable security level and lack of error propagation, most users appear to favour the stream cipher. The synchronization of such a system will be discussed later, after considering the sequence generator algorithm.

The properties of the algorithm should satisfy three requirements. First, the sequence should have a long period. It has to be reproduced at the receive end, where precisely the same sequence must be recovered from the same key. This means that the algorithm calls for what is known as a finite state machine, and a theory in mathematics states that if the input sequence to a finite state machine is ultimately periodic then the output sequence must also be ultimately periodic. Now with our algorithm the input is simply the key, which of course must be ultimately periodic. Thus the sequence produced must also be ultimately periodic, even though it may take a very long time to come back to the starting number and so complete a cycle. It must in fact be guaranteed that the length of this cycle (i.e. the period length) is long, otherwise security will be reduced.

The second requirement is that the sequence must exhibit good randomness properties. The reason is that if the enciphered data stream should be intercepted, the first thing the interceptor may try to do is to perform some kind of statistical analysis on the data. To prevent this

giving him any information, the enciphered data stream must appear perfectly random. Digitized speech cannot be relied upon to provide these good random conditions. Therefore, for the enciphered data stream to appear random, the pseudo-random sequence that is being generated must also appear to be random. In this context there are two types of randomness to consider. What is known as 'global' randomness takes account of the whole of the cycle being generated, and an attempt is made to prove mathematical results about the randomness; for instance counting how many zeros and how many ones occur; trying to find expressions for functions like the autocorrelation function, and so on. In practice however all the sequences that would be used have extremely long periods and it is very uncommon ever to find one less than say 10^{10} bits in length. Thus any particular message that is enciphered will only use a very small portion of a sequence. It is therefore also necessary to look at 'local' randomness; in other words, to take 'small' samples of the sequence and perform a statistical analysis.

The third requirement to be noted is that, given part of the sequence, it should not be possible to discover the key. To appreciate the importance of this, consider the position of the interceptor. Having intercepted some of the enciphered data, he could perhaps predict some of the digitized speech. The response of most common a-d converters to certain signals is known, since they too are finite state machines. Thus suppose a speaker's voice was going through an a-d converter and there was an air conditioning whine in the background. At each pause in the speech, the whine is converted into a number of digits which may well be predictable. Thus there may be many instances when the interceptor actually knows part of the digitized stream that is being enciphered. If he knows both this data stream and the ciphertext, then, by doing a simple modulo-2 addition between the two of them, he can obtain part of the sequence. It is therefore important to be sure that if he has obtained part of the sequence, he still cannot get back to discover the key and therefore cannot predict any more of the sequence.

Each bit of the sequence can be represented as an equation whose variables are the key; the only input to the algorithm is the key and the output is the sequence. If those equations are linear there is a general solution. If they are non-linear then mathematically no general solution is known at present. This is not to say they cannot be solved but means that there is no general solution for them all. In any particular instance these equations must be actually generated and examined to determine whether they can be solved or not. Mathematically this is perhaps one of the most interesting parts of the system.

Turning to the synchronization system, there are essentially two types of synchronization that could be used: initialization-only and continuous. Initialization-only synchronization is self-explanatory and consists simply of inserting a block of synchronization information at the start of the spoken message. Once the other machine receives that information it will 'flywheel' in following the transmission. Continuous synchronization entails sending updates of synchronization information at various intervals throughout the message. Continuous synchronization has advantages, the first of which is that it allows automatic clear-to-secure switching on receive. In practice the synchronization information is only needed for a secure message and not for a clear one. Therefore, at the receive end it is only necessary to look for whether synchronization is present or not. If it is present the message is secure and if not the 'clear' mode is switched in.

Another advantage of continuous synchronization is that it allows late entry. Suppose a net of say 2 or 3 people is already communicating and that at some point during the conversation somebody else wishes to join in. All he would need do is wait for the next update of synchronization information and then join the net. Clearly as far as the user is concerned these are both very important advantages. A possible disadvantage is that if the synchronization information is sent at regular intervals it is then susceptible to jamming. For instance, a pulse jammer could readily destroy the synchronization whenever it is sent. Thus one of the requirements for continuous synchronization must be that it be sent at pseudo-random intervals, preferably depending on the key. It should also be pseudo-random itself, so that not only will an interceptor not know when it is going to appear in a stream of enciphered text but also he will be unable to recognize it when it does appear since it is itself enciphered using the key.

The synchronization must be receivable even over very poor circuits. For instance, even with a 1 in 6 mean bit error rate, 16 kbit/s digitized speech can still be understood. If the range of the radio is not to be reduced, it must be absolutely certain, or there must be a very high probability, that the synchronization signal can be received even over such noisy channels. An example of a system that uses this technique is shown in Fig. 9. As with the earlier example of an analogue scrambler, an attraction is the small size.

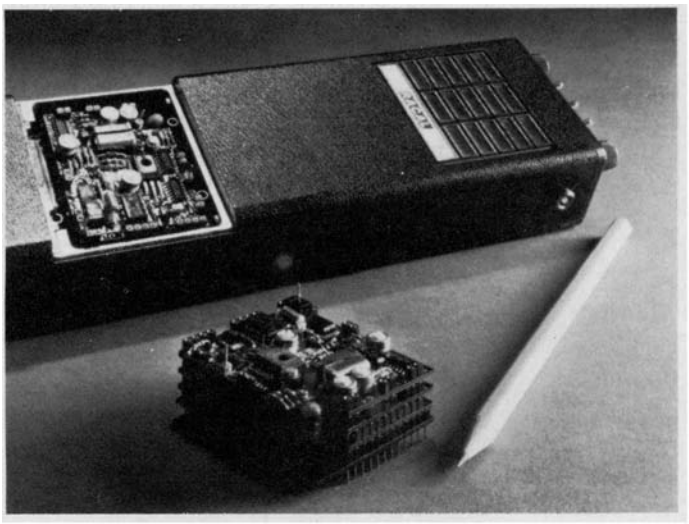


Fig. 9. Racal-Comsec MA4263 with keyfill unit

Again a keyfill device is included for ease of management. The keys are entered into the device and can then be transferred to the MA4263. The MA4263 employs a stream cipher and has approximately 5×10^{27} key settings available. The particular key setting chosen will select one of about $4{\cdot}5 \times 10^{15}$ different sequences, each of which has a period in excess of 10^{12} bits. The synchronization is continuous, sent at pseudo-random time intervals and is itself pseudo-random under the control of the key setting.

5 Discussion

As part of this discussion, an attempt will be made to answer the two most difficult questions that are normally posed when considering speech encryption systems: how secure are they, and where do they fit in, in respect of their security levels, with the other techniques available today? Answers to these questions may well be open to criticism and may be the cause of much debate, but the attempt will be made because of their importance. A difficulty here is

Table 1. Speech security systems (graded)

Inverter systems (no keys)
Frequency bandscrambler—fixed permutation
Frequency bandscrambler—changing permutations
Time element scrambler—fixed permutation
Sliding window system—e.g. MA4258
Time element scrambler—changing permutations
Digital speech cipher—cipher feedback
Digital speech cipher—e.g. MS4263

that any particular system has to be studied in relation to the precise parameters it uses and exactly how the algorithm works. In attempting to produce a realistic grading of systems it must be assumed that in each case the best use of the available system or technique has been made. Thus in comparing systems using sequence generators it is assumed that all such systems use the same sequence generator and that it is a good one. The same applies to permutation generators and so on. It should be pointed out however that in any particular circumstance the order given might change.

Starting at the bottom end with inverter systems these simply take the frequency band of the signal and invert it. Normally there is no key involved and they can be regarded as privacy devices. Next up the list come the common frequency band scramblers which function simply in the frequency domain with a small number of sub-bands; first those that use the same fixed permutation all the time and then those having the improvement of a changing permutation (every so often during the transmission they change the permutations used). Next come the time element scramblers, again with fixed and variable permutation, the latter changing the permutation every frame. Finally there are the digital speech ciphers, beginning with cipher feedback and concluding with the stream cipher technique. The grading is shown in Table 1.

To reinforce the comments made above regarding particular systems, it might for instance be found that a good time element scrambler should be graded higher than a less good digital speech cipher feedback system. The discussion assumes that the best possible use has been made of the technique available.

With regard to the two new options that have been described, the sliding window technique (e.g. the MA4258) is really a medium-security level system, but fits in somewhere around the two time-element scramblers; it is better than a fixed permutation, but almost certainly not quite as good as a system where the permutation will change every frame. The digital stream system, e.g. the MA4263, stands at the highest security level. As a final comparison of the two systems, and in particular the two examples of the MA4258 and the MA4263, a major advantage of the MA4258 is that it preserves the speech bandwidth, in other words it does not increase the bandwidth in any way. Thus it can be used, for instance, on telephone lines and it provides a medium level of security at a reasonable cost. The major advantage of the MA4263 is that it offers the highest level of security and would be the first choice where a high security level was paramount. The two equipments are comparable in size as shown in Figs. 7 and 9, and they are also comparable in audio quality although that is to some extent a subjective consideration.

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